

Note on Data Encryption
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– 1 – Introduction.

Modern computer science relies a lot on security, either via physical solutions or software ones. In this notes we focus on the latter; we mainly cover the most commonly used algorithm in cryptography, as well as some codes used for communications.

In the remainder of this section we provide the reader with some basic definitions of abstract algebra, following with some notion of projective geometry, as these will be at the core of our discussion.

– 1.1 – Notions of abstract algebra.

In this section, we first introduce the notion of group, ring and field, we then define what a field extension is and how to it is used. In describe what above, we assume the reader has a basic understanding of modulo arithmetics.

Definition (Group): Let G be a set, and let $*$ denote a binary operation on G 's elements. We call *group* the pair $(G, *)$ if the following hold:

1. There exists a neutral element in G ; that is,

$$\exists e \in G \text{ s.t. } \forall a \in G, a * e = e * a = a$$

2. Each element has an inverse, i.e.,

$$\forall a \in G, \exists a^{-1} \text{ s.t. } a * a^{-1} = a^{-1} * a = e$$

3. The operation $*$ is associative, meaning that

$$\forall a, b, c \in G, (a * b) * c = a * (b * c)$$

Additionally, if $*$ is such that $a * b = b * a$, i.e., $*$ is commutative, we say that the group is *abelian*.

The number of elements of a group G is said to be its order $|G|$.

Definition (Cyclic group): Let $(G, *)$ be a group, and let $a \in G$. We denote by a^n the n -th power of a , and denote the set of all such powers by $\langle a \rangle$. We say that $(G, *)$ is cyclic if $\langle a \rangle = G$.

Definition (Sub-group): Given $(G, *)$ a group, we say that $H \subseteq G$ forms a sub-group under $*$, if $(H, *)$ is itself a group.

The following is one of the core theorems in group theory, and its extensions to rings, comes in handy when dealing with fields.

Theorem (Lagrange's theorem): Let $(G, *)$ be a group and let $(H, *)$ be a subgroup of G . Then, it holds that $|H|$ divides $|G|$.

Section 1 – Introduction

Groups are one of several algebraic structure, among which we also consider rings and fields. The latter will be at the center of our discussion on cryptography.

Definition (Ring): Let R be a set, and let $+, *$ be two distinct binary operations on R 's elements. We say that $(R, +, *)$ is a ring if

1. $(R, +)$ is an abelian group;
2. The operation $*$ is associative;
3. The operation $*$ distributes over $+$; that is,

$$\forall a, b, c \in R, a * (b + c) = a * b + a * c.$$

Additionally, if R has a multiplicative unit, i.e., $\exists e' \in R$ such that $e' * a = a * e' = a, \forall a \in R$ and $*$ is commutative, we say that $(R, +, *)$ is a commutative ring with unity.

In what follows we denote by $\mathcal{K}^*$ the ring $\mathcal{K} \setminus \{0\}$.

Definition (Field): Given a commutative ring with unity $(R, +, *)$, we say that $(R, +, *)$ is a field if $(R^*, +, *)$ is commutative.

Similarly for groups, each field has an order given by the number of its elements. In these notes we mostly consider fields of order a prime and denote these by $\text{gf}(p)$, for some prime number p .

– 1.2 – A brief introduction to elliptic curves.

We now introduce the notion of *elliptic curve* over a field $\mathcal{K}$, as well as few rules to add two points P, Q belonging to the curve.

Definition (Elliptic curve): Let $\mathcal{K}$ be a field. We call elliptic curve E over $\mathcal{K}$ the set of solution of an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_5$$

such that $a_1, a_2, \dots, a_5 \in \mathcal{K}$ and $\Delta \neq 0$, where Δ is defined as follow:

$$\begin{cases} \Delta = -d_1^2d_4 - 8d_2^3 - 27d_3^2 + 9d_1d_2d_3 \\ d_1 = a_1^2 + 4a_2 \\ d_2 = 2a_4 + a_1a_3 \\ d_3 = a_3^2 + 4a_5 \\ d_4 = a_1^2a_5 + 4a_2a_5 - a_1a_3a_4 + a_2a_3^2 - a_4^2 \end{cases}$$